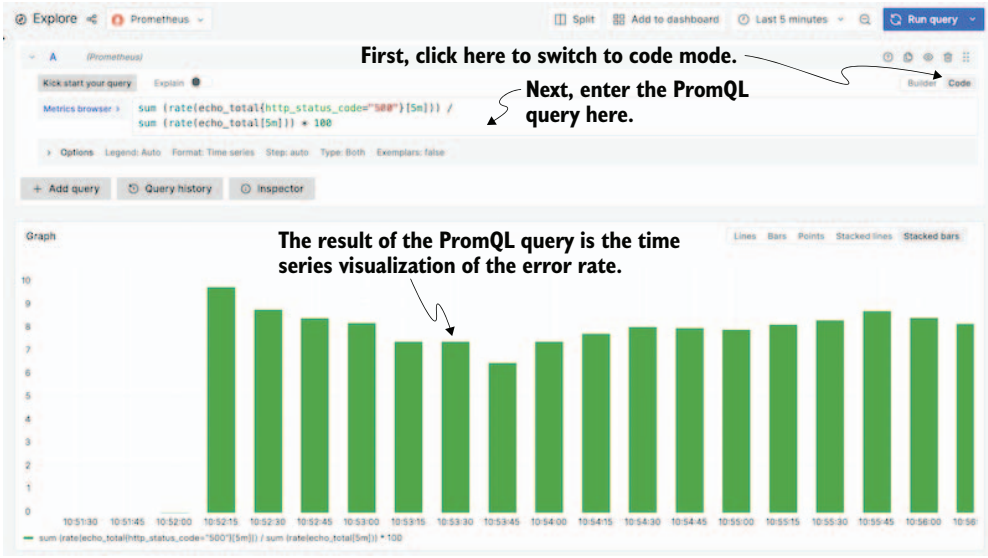


Figure 2.4 Exploring metrics with Prometheus

The preceding PromQL query is the canonical way to capture the error rate. By dividing the errors (HTTP status code 500) by all of the data points, you have a flexible yet complete way to capture errors. In this case, the error rate is a low single-digit rate and should, on average, be 10%, since that's what we have the service configured with in listing 2.7. Next, let's discuss the pros and cons of metrics.

2.4.3 Costs and benefits

We now have a closer look at the costs and benefits of the metrics signal type. Table 2.2 shows an overview of costs associated with metrics.

Table 2.2 Overview of the costs of metrics

	Overhead	Considerations
Instrumentation	Auto-instrumentation, if possible	Increasingly supported in programming languages
Agent	Prometheus, OpenTelemetry collector, and OpenMetrics as emerging standard	What to collect and autopruning
Destination	Cardinality, long-term storage, and federation	What to ingest and data temperature for retention